Advanced IoT Systems & Design: Course Outline

Course Information

Item	Detail
Course title	Advanced IoT Systems & Design
Course code	To be confirmed by the Department
Programme	MSc / MPhil Computer Engineering
Duration	14 weeks (11 lectures over 12 teaching weeks, then 2 class project weeks), followed by examinations
Lecturer	Dr. Dennis A. N. Gookyi
Email	dennisgookyi@gmail.com
Mode of delivery	Lectures (L01 to L11), hands-on labs, paper seminars and a class project

Course Description

This course teaches students to design and build complete Internet of Things (IoT) systems on low-cost microcontrollers, with machine learning running on the device itself. It extends the TinyML approach of from a single smart device to a connected, secure and managed system.

Students build an IoT stack in stages. An Arduino Nano 33 BLE Sense acts as the intelligent sensor node, an ESP32 acts as the gateway to Wi-Fi and the cloud, and data flows through MQTT to a dashboard. Along the way they study IoT architectures, sensing and low-power design, wireless connectivity and protocols, deep learning for microcontrollers, the Edge Impulse platform, IoT security with lightweight cryptography, and cloud integration.

The course is hands-on throughout. It includes a voice-controlled IoT actuator project and ends with a class project on anomaly detection in IoT systems, using examples from agriculture, health and energy in Ghana.

Expected Learning Outcomes

On successful completion of this course, students will be able to:

1. **LO1:** Design a multi-layer IoT architecture (device, gateway, cloud) for a given application and justify its trade-offs in power, cost, latency and reliability.
2. **LO2:** Interface sensors and actuators with low-cost microcontrollers and estimate the power budget and battery life of a sensor node.
3. **LO3:** Connect devices using BLE, Wi-Fi and LoRa, and exchange data with MQTT and CoAP through a gateway.

4. **LO4:** Explain the machine learning paradigm, train and evaluate DNN and CNN models, and interpret metrics such as accuracy, precision, recall and the confusion matrix.
5. **LO5:** Use Edge Impulse to collect data, design an impulse, quantize a model and deploy it for on-device inference on an Arduino Nano 33 BLE Sense.
6. **LO6:** Secure an IoT system through threat modelling, TLS and lightweight cryptography (ASCON).
7. **LO7:** Integrate, test and evaluate an end-to-end IoT system with on-device intelligence, and report it in the style of a research paper.

Textbooks and Course Site

Main textbooks

1. Gian Marco Iodice: *TinyML Cookbook* (2nd ed.), Packt, 2023.
2. Simon Monk: *Programming Arduino: Getting Started with Sketches* (3rd ed.), McGraw Hill, 2022.
3. Perry Lea: *IoT and Edge Computing for Architects* (2nd ed.), Packt, 2020.

Supplementary texts and standards

1. Pete Warden and Daniel Situnayake: *TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers*, O'Reilly, 2019.
2. David Hanes et al.: *IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things*, Cisco Press, 2017.
3. NIST SP 800-232 (2025): *Ascon-Based Lightweight Cryptography Standards for Constrained Devices*.
4. P. Warden (2018): Speech Commands: A Dataset for Limited-Vocabulary Speech Recognition, arXiv:1804.03209.

Course site. Lecture slides, Jupyter notebooks, Arduino sketches and lab templates: https://github.com/dennisgookyi/Advanced-IoT-Systems-and-Design/

Course Hardware and Software

Each group of three or four students works with one kit built around the Arduino Nano 33 BLE Sense, the board, plus an ESP32 gateway that adds Wi-Fi.

Role	Hardware	Why it is used
Intelligent sensor node	Arduino Nano 33 BLE Sense Rev2 (nRF52840, Arm Cortex-M4F, 1 MB flash, 256 KB RAM)	Built-in IMU, microphone, temperature, humidity, pressure and light sensors; BLE; runs TinyML models
Gateway	ESP32 development board	Bridges BLE to Wi-Fi and MQTT; runs FreeRTOS; supports OTA updates
Actuator	28BYJ-48 stepper motor with ULN2003 driver board and external 5 V supply	Voice-controlled actuator project
Extra sensing	Capacitive soil moisture sensor, INA219	Calibration and power

Role	Hardware	Why it is used
	current sensor	measurement labs
Optional	SX1276 LoRa modules; OV7675 camera (as in the Arduino Tiny Machine Learning Kit)	Long-range links; vision projects

Software. Arduino IDE 2 and Arduino CLI; Edge Impulse Studio and Edge Impulse CLI; Python 3 with Jupyter Notebook or Google Colab, TensorFlow and TensorFlow Lite; Mosquitto MQTT broker; Node-RED, InfluxDB and Grafana (via Docker); Git and GitHub.

Schedule

Lecture	Topic	Week	Lab / milestone
L01	Course Overview and IoT System Architectures	1	Lab 0 (self-study)
L02	Course Hardware and Software Toolchain Setup	2	Lab 1: Toolchain and on-board sensors
L03a	Sensing and Data Acquisition	3	Lab 2: Sensor data logging and calibration; groups formed
L03b	Low-Power Design for IoT Nodes	4	Lab 3: Power measurement and battery life; project proposal due
L04a	Wireless Connectivity: BLE, Wi-Fi and LoRa	5	Lab 4: BLE sensor node and ESP32 gateway
L04b	IoT Protocols and Gateway Firmware: MQTT, CoAP, FreeRTOS	6	Lab 5: Gateway to MQTT broker
L05	Overview of Tiny Machine Learning for IoT	7	Lab 6 (part 1): ML in Jupyter; mid-semester test
L06	Overview of Deep Learning for Microcontrollers	8	Lab 6 (part 2): DNN, CNN and quantization
L07	Overview of the Edge Impulse Platform	9	Lab 7: Motion classifier on the Nano; project progress review
L08	Hands-on Project: Voice-Controlled IoT Actuator	10	Lab 8: Keyword spotting drives a stepper and reports over BLE
L09	IoT Security and	11	Lab 9: TLS and ASCON

Lecture	Topic	Week	Lab / milestone
	Lightweight Cryptography		
L10	Cloud, Dashboards and Device Management	12	Lab 10: Dashboard, OTA and digital twin; paper seminars end
L11	Class Project: Advanced Anomaly Detection in IoT Systems	13	Supervised project work
	Project presentations and demonstrations	14	Demonstrations; final report due

Assessment Summary

Component	Weight	Timing	LOs assessed
Laboratory reports (Labs 1 to 10)	10%	Weekly, due one week after each lab	LO2 to LO6
Research paper seminar	5%	Weeks 5 to 12	LO1, LO7
Mid-semester test	5%	Week 7 (L01 to L04b)	LO1 to LO3
Class project: Advanced Anomaly Detection in IoT Systems	20%	Proposal Week 4, review Week 9, demo and report Week 14	LO1 to LO7
Final examination (3 hours)	60%	Examination period	LO1 to LO6
Total	**100%**		

Useful Links

- https://www.tensorflow.org/lite
- https://www.edgeimpulse.com/
- https://docs.edgeimpulse.com/
- https://www.arduino.cc/
- https://docs.arduino.cc/hardware/nano-33-ble-sense-rev2/
- https://docs.espressif.com/projects/arduino-esp32/
- https://mosquitto.org/
- https://ascon.isec.tugraz.at/
- https://tinyml.seas.harvard.edu/
- https://www.adafruit.com/